

Shenglai Zeng

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Research Interest

I am a fourth-year Ph.D. candidate at Michigan State University, advised by Professor Jiliang Tang, and I am on the job market for positions starting in 2027. My research builds **personalized, proactive, and safe agents**. I work primarily on the **agent harness** — the self-improvement, memory, retrieval, and evaluation/environment machinery that determines what an agent can actually do — and extend it to **multimodal interaction** and **post-training**. I have published 35+ papers at NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, and CVPR, received the Best Paper Award of IEEE Transactions on Cloud Computing (2023) and an ACL-2026 SAC Highlight (top 1%, Best Paper Candidate), and have industry research experience at Meta AI, Amazon, and Baidu.

Education

Michigan State University, East Lansing, USA

Sept. 2023 – Present

Ph.D. Candidate in Computer Science and Engineering

- ▷ Advisor: Dr. Jiliang Tang | Lab: Data Science and Engineering (DSE) Lab
- ▷ Expected Graduation: **May 2027**
- ▷ Research Focus: Agentic AI, Retrieval & Memory, Personalized LLMs, Multimodal Modeling, Trustworthy AI

University of Electronic Science and Technology of China, Chengdu, China

Sept. 2019 – Jun. 2023

B.Sc. in Computer Science and Engineering, Yingcai Honor School

- ▷ CGPA 3.98/4.00; Weighted Average 93.97/100 (**1st among 100 students**)

Honors and Awards

- ▷ **SAC Highlight (Top 1%) & Best Paper Candidate**, ACL-2026 2026
- ▷ **Best Paper Award**, IEEE Transactions on Cloud Computing (JCR Q1) 2023
- ▷ **The Most Outstanding Students Award**, UESTC (highest honor; 10 students university-wide)
- ▷ **National Scholarship** (highest undergraduate honor in China) 2020
- ▷ WAC Scholarship (10 undergraduates university-wide per year), UESTC 2021
- ▷ 1st Outstanding Academic Scholarship (Top 5%), UESTC 2020, 2021, 2022

Industry Experience

*All papers in this section are **first-authored** (including co-first).*

Meta AI, Seattle, USA

Summer 2026

Research Scientist Intern — Next Generation Proactive Agents

Mentored by Xiao Yang, [Luna Dong](#), and [Scott Yih](#); closely worked with a cross-organizational team spanning MSL, Reality Labs, FAIR, and MRS.

I built the **memory layer for proactive agents** — assistants that decide *whether and when* to speak, not just what to answer. Existing systems fail outright in this regime: on **MPA**, a benchmark I built that pairs egocentric video with year-long, distractor-rich personal memories (~**618K tokens** each) and scores content and timing jointly, **every frontier model scores near zero**. My plug-and-play harness **GPM** surfaces memory from

a graph-organized store and gates it down to what is worth interrupting for, lifting strict accuracy to **37.0%** (Qwen3.8-27B) and **38%** on GPT-5.3-Chat with no training, reaching **42.8%** with **RL post-training**.

▷ Memory-Driven Proactive Agents [Submitted to ICLR-2027].

Amazon, Search & Shopping FM Team, CA, USA

Summer 2024, 2025

Applied Scientist Intern — Retrieval-Augmented Generation and Personalization

Mentors: Tianqi Zheng, Dante Everaert | Managers: Hanqing Lu, Chuan Tian, Monica Xiao Cheng

I worked on what it takes to ship retrieval-augmented and personalized LLMs: cutting personalization context by **50×** (10K→200 tokens) at near-full-context quality [ACL-2026], making models context-robust (+26–35 points over the base LLM) while tuning only **0.0004% of parameters on fewer than 200 examples** [ACL-2025], and filtering misleading retrieved evidence at **95% accuracy** [NAACL-2025]. On the data side, I replaced private retrieval corpora with agent-refined **synthetic data** at comparable task quality [EMNLP-2025], and ran the **first privacy audit of multimodal RAG**, where **277 of 500** attack prompts extracted retrieved image-text content [EMNLP-2025]. Invited back for a second summer.

- ▷ Efficient Personalized LLMs (PLLMs) [ACL-2026, Oral, Best Paper Candidate, SAC Highlight].
- ▷ Knowledge Checking in RAG [NAACL-2025, Oral].
- ▷ Representation Finetuning for Robust RAG [ACL-2025].
- ▷ Self-Improving Synthetic Data for RAG [EMNLP-2025].
- ▷ Multi-modal RAG [EMNLP-2025].

Baidu Inc., Search & Agent Business Team, Beijing, China

May 2023 – May 2024

Research Intern — RAG Privacy and LLM Memorization

Mentors: Dr. Yiding Liu, Dr. Dawei Yin

I established the privacy foundations of retrieval-augmented generation: the **first work to reveal that retrieval-augmented systems carry severe privacy risk**, leaking their own retrieval database [ACL-2024]. The threat model it introduced has since been carried into the privacy analysis of future systems — multi-modal RAG, agent memory, and multi-agent systems — and the paper now has **350+ citations**. I also produced the **first task-level map of memorization in LLM post-training** [ACL-2024], showing how memorization **scales with model size** and that the scaling itself is task-dependent (22.3% for summarization vs. 0.15% for reading comprehension).

- ▷ RAG Privacy and Retrieval-Database Leakage [ACL-24].
- ▷ Memorization in LLM post-training [ACL-2024].

Publications

35+ papers at top-tier ML/NLP venues (NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, CVPR), including **12 first-authored** publications. **1340+ citations** (Google Scholar, Sept. 2026). Recognitions include a **Best Paper Award** (IEEE TCC) and **Best Paper Candidate** (ACL), **Spotlight** (ICLR), **Oral** presentations (ACL, NAACL), and adoption in frontier-model technical reports (Muse Spark, Qwen, Kimi). * denotes equal contribution.

First-Authored Publications

- [NeurIPS'26] **S. Zeng**, Q. Wang, K. Guo, X. Dai, X. Long, H. Liu, “Magnifying What Matters: Attention-Guided Adaptive Rendering for Visual Text Comprehension”, *Conference on Neural Information Processing Systems*, 2026.
- [ICML'26] **S. Zeng***, J. Zhang*, K. Guo, X. Dai, H. Liu, J. Tang, Y. Chang, “Fix Before Search: Benchmarking Agentic Query Visual Pre-processing in Multimodal Retrieval-Augmented Generation”, *International Conference on Machine Learning*, 2026.
- [ACL'26, Oral] **S. Zeng**, T. Zheng, C. Tian, D. Everaert, Y.-S. Wang, Y. Huang, M. J. Morais, R. Patki, J. Tian, X. Dai, K. Guo, M. X. Cheng, H. Liu, “Attn-GS: Attention-Guided Context Compression for

Efficient Personalized LLMs”, *Annual Meeting of the Association for Computational Linguistics*, **Oral, SAC Highlight (Top 1%), Best Paper Candidate**, 2026.

[EMNLP’25] **S. Zeng**, J. Zhang, P. He, J. Ren, T. Zheng, H. Lu, H. Xu, H. Liu, Y. Xing, J. Tang, “Mitigating the Privacy Issues in Retrieval-Augmented Generation (RAG) via Pure Synthetic Data”, *Conference on Empirical Methods in Natural Language Processing*, 2025.

[EMNLP’25] **S. Zeng***, J. Zhang*, J. Ren, T. Zheng, H. Liu, X. Tang, Y. Chang, “Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation”, *Conference on Empirical Methods in Natural Language Processing*, 2025.

[ACL’25] **S. Zeng**, P. He, K. Guo, T. Zheng, H. Lu, Y. Xing, H. Liu, “Towards Context-Robust LLMs: A Gated Representation Fine-tuning Approach”, *Annual Meeting of the Association for Computational Linguistics*, 2025.

[NAACL’25, Oral] **S. Zeng**, J. Zhang, B. Li, Y. Lin, T. Zheng, D. Everaert, H. Lu, H. Liu, Y. Xing, M. X. Cheng, J. Tang, “Towards Knowledge Checking in Retrieval-Augmented Generation: A Representation Perspective”, *Conference of the North American Chapter of the Association for Computational Linguistics*, **Oral**, 2025.

[ACL’24] **S. Zeng***, Y. Li*, J. Ren, Y. Liu, H. Xu, P. He, Y. Xing, S. Wang, J. Tang, D. Yin, “Exploring Memorization in Fine-tuned Language Models”, *Annual Meeting of the Association for Computational Linguistics*, 2024.

[ACL’24] **S. Zeng***, J. Zhang*, P. He, Y. Xing, Y. Liu, H. Xu, J. Ren, S. Wang, D. Yin, Y. Chang, J. Tang, “The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)”, *Findings of the Association for Computational Linguistics: ACL*, 2024.

[IEEE TCC, Best Paper] **S. Zeng**, Z. Li, H. Yu, Z. Zhang, L. Luo, B. Li, D. Niyato, “HFedMS: Heterogeneous Federated Learning with Memorable Data Semantics in Industrial Metaverse”, *IEEE Transactions on Cloud Computing*, **Best Paper Award**, 2023.

[DASFAA’22, Oral] **S. Zeng**, Z. Li, H. Yu, Y. He, Z. Xu, D. Niyato, H. Yu, “Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training”, *International Conference on Database Systems for Advanced Applications*, **Oral**, 2022.

[SDM’23] **S. Zeng***, J. Wang*, Z. Long, Y. Wang, H. Xiao, F. Ma, “Knowledge-Enhanced Semi-Supervised Federated Learning for Aggregating Heterogeneous Lightweight Clients in IoT”, *SIAM International Conference on Data Mining*, 2023.

Co-Authoring Publications

[NeurIPS’26] Y. Li, Y. Feng, Z. Xu, Z. Ma, K. Zheng, F. Jiang, X. Sun, R. Shao, Z. Chen, Y. Huang, X. Han, B. Lee, K. Xu, **S. Zeng**, H. Hua, X. Zhang, B. Alomair, R. Krishna, L. Zettlemoyer, P. W. Koh, B. Ramasubramanian, L. Niu, X. Yue, R. Poovendran, “JobBench: Aligning Agent Work With Human Will”, *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2026. (adopted by Meta Muse Spark 1.1, Moonshot Kimi K3, and Alibaba Qwen 3.8 Max)

[EMNLP’26] K. Guo, X. Dai, **S. Zeng**, H. Shomer, H. Han, Y. Wang, J. Tang, “Beyond Static Retrieval: Opportunities and Pitfalls of Iterative Retrieval in GraphRAG”, *Conference on Empirical Methods in Natural Language Processing*, 2026.

[ICML’26] X. Dai, K. Yang, C. Luo, **S. Zeng**, K. Guo, J. Tang, “When Do Hallucinations Arise? A Graph Perspective on the Evolution of Path Reuse and Path Compression”, *International Conference on Machine Learning*, 2026.

[KDD’26] K. Guo, X. Dai, Z. Zhang, N. Lin, **S. Zeng**, J. Ren, H. Han, J. Tang, “Why Retrieval-Augmented Generation Fails: A Graph Perspective”, *ACM SIGKDD Conference on Knowledge Discovery*

and Data Mining, 2026.

- [NeurIPS'25] J. Ren, Z. Dai, X. Tang, Y. Xing, **S. Zeng**, H. Liu, J. Zeng, Q. Peng, S. Varshney, S. Wang, Q. He, C. C. Aggarwal, H. Liu, "Keeping an Eye on LLM Unlearning: The Hidden Risk and Remedy", *Conference on Neural Information Processing Systems*, 2025.
- [EMNLP'25] K. Guo, H. Shomer, **S. Zeng**, H. Han, Y. Wang, J. Tang, "Empowering GraphRAG with Knowledge Filtering and Integration", *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [ACL'25] B. Wang, W. He, P. He, **S. Zeng**, Z. Xiang, Y. Xing, J. Tang, "Unveiling Privacy Risks in LLM agent memory", *Annual Meeting of the Association for Computational Linguistics*, 2025.
- [KDD'25] B. Li, Z. Chen, H. Han, **S. Zeng**, J. Liu, J. Tang, "Unveiling Mode Connectivity in Graph Neural Networks", *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025.
- [CVPR'25] J. Ren, K. Chen, Y. Cui, **S. Zeng**, H. Liu, Y. Xing, J. Tang, L. Lyu, "Six-CD: Benchmarking Concept Removals for Benign Text-to-Image Diffusion Models", *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [TMLR] P. He, Y. Xing, H. Xu, J. Ren, Y. Cui, **S. Zeng**, J. Tang, M. Yamada, M. Sabokrou, "Stealthy Backdoor Attack via Confidence-driven Sampling", *Transactions on Machine Learning Research*.
- [ECCV'24] J. Ren, Y. Li, **S. Zeng**, H. Xu, L. Lyu, Y. Xing, J. Tang, "Unveiling and Mitigating Memorization in Text-to-Image Diffusion Models through Cross Attention", *European Conference on Computer Vision*, 2024.
- [EMNLP'24] H. Xu, J. Ren, P. He, Y. Cui, **S. Zeng**, H. Liu, J. Tang, A. Liu, "On the Generalization of Training-based ChatGPT Detection Methods", *Findings of the Association for Computational Linguistics: EMNLP*, 2024.
- [ICLR'24, Spotlight] P. He, H. Xu, J. Ren, Y. Cui, **S. Zeng**, H. Liu, C. Aggarwal, J. Tang, "Sharpness-Aware Data Poisoning Attack", *International Conference on Learning Representations*, **Spotlight**, 2024.
- [NeurIPS'23] J. Li, H. Shomer, H. Mao, **S. Zeng**, Y. Ma, N. Shah, J. Tang, D. Yin, "Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking", *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2023.

Preprints and Under Review

- [Preprint] X. Long, Z. Chen, **S. Zeng**, S. Wang, K. Guo, J. Tang, "MemTrace: Probing What Final Accuracy Misses in Long-Term Memory".
- [Preprint] X. Dai, C.-H. Lo, K. Guo, **S. Zeng**, D. Luo, J. Tang, "Uncovering Graph Reasoning in Decoder-only Transformers with Circuit Tracing".
- [Under review] Z. Chen, J. Gu, J. Yin, X. Long, **S. Zeng**, X. Liu, K. Guo, K. Zhou, J. Tang, "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline", *ACL Rolling Review*.
- [Under review] X. Dai, K. Guo, C.-H. Lo, **S. Zeng**, J. Ding, D. Luo, S. Mukherjee, J. Tang, "GraphGhosts: Tracing Reasoning Structures Behind Large Language Models", *ACL Rolling Review*.
- [Preprint] Y. Lin, P. He, H. Zhao, **S. Zeng**, Z. Xiang, H. Liu, J. Tang, "Structural Alignment Faking: Hiding Malicious Capabilities in Mixture-of-Experts LLMs".
- [Preprint] J. Ren, H. Xu, P. He, Y. Cui, **S. Zeng**, J. Zhang, H. Wen, J. Ding, H. Liu, Y. Chang, J. Tang, "Copyright Protection in Generative AI: A Technical Perspective".

Community Services

- ▷ **Conference Reviewer / Program Committee:** NeurIPS, ICML, ICLR; ACL, EMNLP, NAACL (via ACL Rolling Review); MICCAI Workshop on Distributed, Collaborative and Federated Learning (DeCaF-2022).
- ▷ **Journal Reviewer:** ACM Transactions on Information Systems (TOIS), IEEE TKDE, IEEE TMC, IEEE TPDS, IEEE Transactions on Information Forensics & Security, IEEE Transactions on Services Computing, IEEE Transactions on Vehicular Technology, IEEE Internet of Things Journal.
- ▷ Reviewed **30+** papers.